Python Basics Cheat Sheet (Module 5)

Tool: Google Colab. All examples use ShopKart values.

Colab essentials

Do this	How
Run a cell	Shift + Enter
New code / text cell	+ Code / + Text (top bar)
Open a course notebook	File → Upload notebook
Keep your work	File → Save a copy in Drive
Add data	Folder icon (left) → drag the CSV in
Clear variables	Runtime → Restart session
Run everything	Runtime → Run all
Warning	Uploaded files disappear when the runtime resets. Upload again.

Variables and types

```
order_id = "O10006"      # str  (text, in quotes)
quantity = 2             # int  (whole number)
discount = 0.05          # float (decimal)
is_returned = False      # bool (True / False)
type(quantity)           # <class 'int'>
revenue = quantity * 1440 * (1 - discount)   # 2736.0
round(revenue, 2)
print(f"Order {order_id}: ₹{revenue}")      # f-string
```

Operators: `+ - * /`, `**` power, `//` whole division, `%` remainder.

Converting and cleaning text

```
int("1299")      # 1299      float("0.05")  # 0.05     str(1440)  # "1440"
int(2.9)         # 2 (chops, does not round)
"₹1,299".replace("₹", "").replace(",", "")   # "1299"
" mumbai ".strip().title()                   # "Mumbai"
"delhi" == "Delhi"                           # False (case-sensitive)
```

Lists (ordered, index from 0)

```
amounts = [1737.6, 570.0, 2145.1, 3434.0]
amounts[0]        # 1737.6 (first)      amounts[-1]   # 3434.0 (last)
amounts[0:2]      # [1737.6, 570.0] (stop not included)
len(amounts), sum(amounts), max(amounts), min(amounts)
amounts.append(3486.6)       # add to end
"Pune" in ["Delhi", "Pune"]  # True
sorted(amounts, reverse=True)
```

Dictionaries (key → value)

```
customer = {"customer_id": "C0593", "name": "Kavya Khan", "age": 26,
"membership": "Prime"}
customer["name"]                  # 'Kavya Khan'
customer["age"] = 27              # update
customer["city"] = "Delhi"        # add
customer.get("phone", "-")        # safe lookup
customer.keys(), customer.values()
```

A list of dictionaries = a mini table: `pd.DataFrame(list_of_dicts)`.

Conditions

```
if revenue >= 5000:
    size = "Big"
elif revenue >= 1000:
    size = "Medium"
else:
    size = "Small"
```

== compare, = store. != > < >= <=. Combine with `and`, `or`, `not`. Colon + 4-space indent.

Loops

```
total = 0
for amount in amounts:
    total += amount               # same as total = total + amount

clean = []
for c in [" Mumbai", "Bombay", "DELHI"]:
    c = c.strip().title()
    clean.append({"Bombay": "Mumbai"}.get(c, c))

for i in range(3): ...            # 0, 1, 2
while stock >= 2: stock -= 2      # make sure it ends!
```

Functions

```
def calc_revenue(quantity, unit_price, discount=0):
    """Revenue of one order."""
    return round(quantity * unit_price * (1 - discount), 2)

calc_revenue(2, 1440, 0.05)   # 2736.0
calc_revenue(1, 2258)         # 2258 (default discount 0)
```

Reading errors (read the LAST line first)

Error	Typical cause	Fix
IndentationError	No indent after `if/for/def`	Indent 4 spaces
SyntaxError	Missing colon, bracket or quote; = instead of ==	Check the line shown
NameError	Typo, or the cell that created it was	Fix spelling / Runtime → Run all

Error	Typical cause	Fix
	not run	
TypeError	`"₹1,299" + 100` (text + number)	Convert first
ValueError	`int("₹1,299")`	Remove ₹ and commas first
KeyError	Missing dict key / column name	Check spelling and capitals
FileNotFoundError	File not in Files panel	Upload again

Libraries and NumPy

```
import numpy as np
import pandas as pd

prices = np.array([362, 300, 2258, 2020])
qty = np.array([6, 2, 1, 2])
prices * qty                  # array([2172, 600, 2258, 4040]) - no
loop
(prices * 1.18).round(2)      # add 18% GST to every price
prices.sum(), prices.mean(), prices.max(), np.median(prices)
prices[prices > 1000]         # filter
(prices > 1000).sum()         # count matches

orders = pd.read_csv("orders_raw.csv")   # (5055, 12)
```

List vs array: `[1, 2] * 2` → `[1, 2, 1, 2]`; `np.array([1, 2]) * 2` → `array([2, 4])`.